

HEREFORD, TX, USA

WMO: 720393

Lat: 34.850N

Lon: 102.333W

Elev: 3783

StdP: 12.79

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	16.0	0.0	6.3	28.6	3.5	7.6	32.5	34.4	54.7	32.1	53.2	8.1	20	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	99.6	64.1	97.2	65.0	94.9	65.4	71.8	85.5	70.5	84.3	69.4	83.3	14.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	119.8	76.2	67.1	114.8	75.1	66.0	110.3	74.3	38.2	85.8	37.0	84.6	36.0	83.1	95.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.5	26.8	24.0	DB	3.1	104.6	6.0	2.8	-1.2	106.6	-4.8	108.2	-8.2	109.8	-12.5	111.8	
			WB	2.0	75.5	5.6	7.1	-2.0	80.6	-5.2	84.7	-8.3	88.7	-12.4	93.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	38.5	41.4	50.4	57.3	66.0	76.9	78.6	77.9	71.2	58.9	47.8	39.4
	DBStd	16.80	9.45	10.89	9.54	8.40	8.44	5.61	4.53	4.91	6.08	9.07	9.38	9.36
	HDD50	1307	370	274	109	30	4	0	0	0	0	29	147	344
	HDD65	3836	823	663	456	250	91	1	0	0	18	225	516	793
	CDD50	4517	12	32	123	250	501	807	887	866	636	306	82	15
	CDD65	1571	0	1	4	20	123	358	422	402	204	37	0	0
	CDH74	20501	2	43	282	713	2060	4507	5089	4701	2343	658	99	4
	CDH80	10428	0	7	65	235	980	2495	2799	2574	1054	210	9	0
Wind	WSAvg	11.0	10.8	11.7	12.1	13.2	12.3	12.0	9.6	9.0	9.4	10.2	10.8	10.8
Precipitation	PrecAvg	19.4	0.6	0.5	1.3	1.1	2.0	3.2	2.3	3.1	1.9	1.6	0.8	0.8
	PrecMax	33.5	1.8	1.4	4.3	5.0	5.7	7.2	5.0	6.7	6.3	7.4	3.6	2.7
	PrecMin	13.7	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.0
	PrecStd	4.5	0.6	0.5	1.1	1.1	1.6	1.9	1.3	1.5	1.5	1.7	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.4	79.4	86.3	90.7	98.0	104.6	103.0	100.5	95.6	89.9	81.4	72.8
		MCWB	48.3	51.7	54.0	55.2	59.4	62.0	65.8	65.0	63.4	60.6	52.4	49.2
	2%	DB	67.9	73.4	81.3	86.1	93.5	100.1	98.9	98.6	93.1	85.7	77.1	67.6
		MCWB	47.0	48.8	51.9	54.4	58.0	62.8	65.7	65.9	63.9	58.9	52.6	47.7
	5%	DB	63.0	68.6	76.8	82.1	90.1	97.0	96.1	95.5	90.2	81.8	71.7	63.0
		MCWB	44.1	46.9	51.1	53.4	57.9	63.9	66.5	66.9	63.7	58.2	50.4	45.7
	10%	DB	57.2	62.6	71.7	78.0	86.0	93.1	93.1	92.7	86.6	77.0	65.9	57.4
		MCWB	41.7	44.4	49.1	53.4	58.1	65.3	67.2	66.7	64.0	57.1	48.9	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	53.6	59.2	62.5	68.7	73.4	74.3	72.9	71.1	69.1	58.5	54.1
		MCDB	68.2	73.1	68.7	73.5	78.5	88.5	87.8	86.5	83.5	73.9	65.8	62.9
	2%	WB	47.6	50.8	56.1	60.0	66.2	70.9	72.0	71.4	69.2	66.1	55.2	49.7
		MCDB	64.3	68.7	70.3	69.8	77.2	85.1	85.3	85.6	80.7	73.1	67.4	63.4
	5%	WB	45.1	48.1	53.2	57.8	64.6	69.4	70.6	70.1	68.0	63.4	52.7	47.1
		MCDB	61.1	67.0	69.6	70.2	76.6	83.4	84.4	84.6	79.4	72.7	68.1	60.3
	10%	WB	42.4	45.3	51.0	55.4	62.6	68.3	69.2	69.1	66.6	60.7	50.3	44.2
		MCDB	56.1	61.9	68.4	72.6	75.2	82.8	83.8	83.6	78.1	70.5	65.1	57.4
Mean Daily Temperature Range	5% DB	MDBR	30.7	30.1	32.7	32.2	30.4	28.6	26.3	27.0	27.0	28.8	30.1	28.1
		MCDBR	37.5	40.3	39.7	39.4	36.9	33.6	30.5	30.3	32.2	34.7	37.8	36.3
		MCWBR	20.1	20.1	18.1	16.2	14.2	9.9	7.6	7.5	9.6	13.4	17.3	19.6
	5% WB	MCDBR	34.9	38.5	34.0	29.1	28.5	26.9	25.0	26.1	24.0	23.5	32.7	33.1
		MCWBR	19.2	20.1	17.4	14.1	13.0	10.2	8.2	7.8	8.4	11.2	16.4	18.9
Clear-Sky Solar Irradiance	taub	0.252	0.262	0.288	0.320	0.334	0.375	0.383	0.368	0.339	0.301	0.272	0.256	
	taud	2.592	2.578	2.507	2.413	2.437	2.363	2.384	2.395	2.450	2.541	2.550	2.577	
	Ebn at Noon	310	317	314	306	301	287	284	287	292	298	298	302	
	Edh at Noon	24	27	32	37	36	39	38	37	33	28	24	23	
All-Sky Solar Radiation	RadAvg	1028	1289	1694	2084	2306	2439	2341	2103	1788	1384	1085	883	
	RadStd	83	108	121	127	160	135	100	99	102	141	102	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	N/A

Nomenclature: See separate page